

MIGRATING WILDEBEEST

In Tanzania's Serengeti National Park, wildebeest migrate in a cyclical path to take advantage of fresh plant growth at different times of the year. They spend the wet season in open grassland, and the dry season in wooded savanna.

**FIRE**

Fires, ignited by lightning, are a natural feature of grassland life, clearing away dead growth, and allowing fresh grass to sprout. In the long run, fires help grassland wildlife, but while they are burning they can be lethal. As the flame front advances, most animals react by running or flying for safety, often abandoning their usual caution in an urgent bid to escape. Some animals, such as bustards and storks, have learned to make the most of this frenzied exodus from the flames. They gather close to fires and snap up insects and other small animals as they scurry away; and once the fire has moved on, they pick over the charred ground for casualties.

**FEEDING BY THE FLAMES**

A European white stork searches for small animals fleeing before an advancing fire. Soaring high over the plains, the stork will have been attracted by the fire's smoke.

America's prairies, some prairie-dog burrow systems covered several thousand square miles and housed millions of animals.

Termites are also accomplished builders, constructing giant, elaborate, subterranean nests that extend high above ground level. These nests house large, cooperative communities that can contain over 30 million inhabitants. Along with ants, they make up a very large part of the habitat's animal life, and provide food for the large insect eaters.

Feeding

Although grass is rich in nutrients and easy to find, it is difficult to digest. Many mammals, including humans, cannot break it down at all because it contains large amounts of cellulose—a carbohydrate that most animals cannot digest. Grazing mammals, however, have special microorganisms in the gut that break down cellulose so that the body is able to use it. Some nonmammal species also use microbes to digest plant material. In tropical savanna, for example, termites rely on them to break down dead leaves and wood.

The animals that are most efficient at using cellulose are ruminant mammals—antelopes, buffaloes, and giraffes, for example—which helps to explain why these animals dominate grasslands. The ruminant's complex stomach acts like a fermentation tank, working to extract the maximum amount of nutrients from food. The animal assists in the process by regurgitating its food and chewing it a second time, making it even easier to break down. Nonruminant plant eaters, such as zebras, have less efficient digestive systems and must therefore eat more to survive.

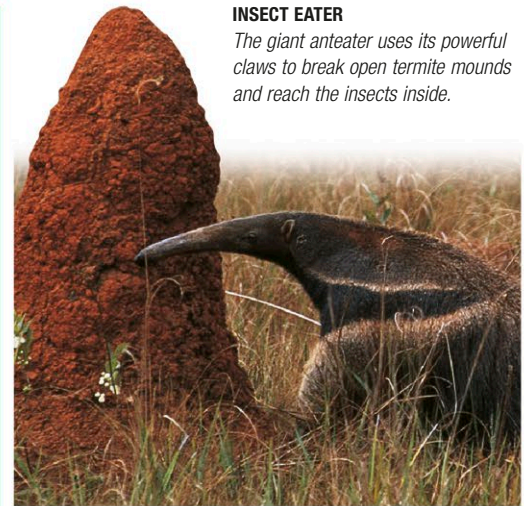
In pure grassland, plant eaters compete for the same food, although each may have a preference for a different type of grass. In savanna, the presence of trees and shrubs makes for a wider range of food, and browsing mammals have minimized competition further by evolving specific ways of feeding. This means that a remarkable number of species can live side by side. For example, the small Kirk's dik dik antelope feeds on shoots and fruit, and rarely touches grass, while the much larger eland will eat almost anything from fruit and seeds to roots scraped up from the ground.

**REACHING FOR FOOD**

Unlike most antelopes, the gerenuk can stand on its back legs. This means that although it measures only about 39 in (1 m) at the shoulder, it can browse leaves over 6½ ft (2 m) up—much higher than the reach of other antelopes of similar size.

INSECT EATER

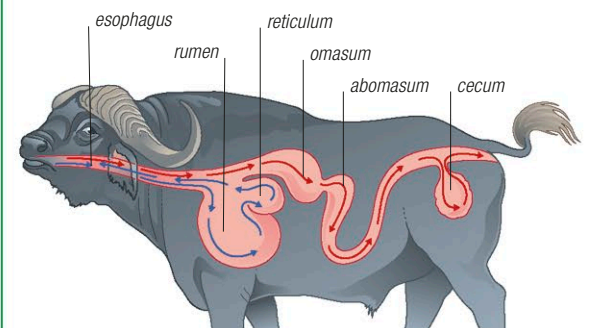
The giant anteater uses its powerful claws to break open termite mounds and reach the insects inside.

**COLLECTING DUNG**

Dung beetles make use of the large amount of dung produced by grazing mammals. They gather the dung into balls, which they then roll away and bury below ground where it is used as food for the beetles' grubs.



Scavengers, which also play an important part in the habitat's ecology, include birds as well as coyotes, jackals, and hyenas. Most airborne scavengers are vultures, but there are several species of scavenging storks, one of which is the marabou stork.

**DIGESTING GRASS**

The largest of the 4 chambers in a buffalo's stomach, the rumen, contains millions of bacteria and protozoans. These simple organisms produce cellulose-splitting enzymes that turn cellulose into simple nutrients that can then be absorbed. The blue arrows show passage of food through the rumen, before regurgitation. Then it is passed on for final digestion, as shown by the red arrows.